Quantum chemical approach for condensed-phase thermochemistry (VI): harmonic solvation model with Gaussian-charge polarizable continuum model

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Abstract

This letter proposes to the Gaussian-charge harmonic solvation model (GC-HSM) for condensed-phase thermochemistry. The conventional point-charge polarizable continuum model (PCM) occasionally causes numerical instabilities in Hessian calculations in the HSM framework due to discontinuous solvation potentials. To overcome this limitation, we implemented analytical first and second derivatives of the solvation energy using a Gaussian-charge discretization scheme. Numerical assessments demonstrate the improved stability of the derivatives and significant computational acceleration compared with numerical differentiation. The present approach enhances the reliability and efficiency of thermodynamic calculations in condensed-phase.

Keywords:

Condensed-phase thermochemistry

Quantum chemistry

Harmonic solvation model

Gaussian-charge scheme

1. INTRODUCTION

Chemical reactions proceed in various environments such as gas phase, liquid phase, solid phase, and on solid surfaces. Thermodynamic quantities governing the direction and equilibrium states of chemical reactions, as well as activation barriers determining the reaction rates are defined by changes in Gibbs or Helmholtz free energy. Although the electronic energy of a system obtained from quantum chemical calculations based on the Born-Oppenheimer approximation accounts for the majority of the free energy, the evaluation of entropy is essential for understanding the temperature dependence of chemical reactions.

In standard gas-phase thermochemistry, thermodynamical parameters are evaluated within the ideal gas model, referred to as IGM, which assumes that molecules behave as an ideal gas. Previously, our group pointed out that the IGM cannot accurately estimate the free energy in solution due to the overestimation of entropy, and proposed an alternative model, referred to as the harmonic solvation model (HSM) [1].

The HSM accounts for solute-solvent interaction within the framework of the polarizable continuum model (PCM) [2,3]. In the HSM framework, the interaction with the cavity is harmonically incorporated as the interaction with the solvent. In thermodynamical calculations, the Hessian matrix is constructed under fixed cavity

conditions. The six (five for linear molecules) lowest frequencies are interpreted as quasi-translational and quasi-rotational modes of a solute bound by the solvent cavity applying the harmonic-oscillator formalism to these modes yields physically meaningful enthalpy and entropy in solution, in contrast to the IGM. In the past, we have applied the HSM to various systems [1,4–7] and demonstrated that quantitative evaluation of condensed systems is possible.

In the HSM formulation, the definition of the PCM cavity is a fundamental aspect. The PCM represents the bulk solvent as a dielectric continuum surrounding the molecular cavity and reduces the computational cost compared to explicit solvation models. In the standard PCM [2] formulations, the cavity surface is typically discretized using the point-charge scheme, where the surface is tessellated into elements carrying apparent charges. However, this approach is not strictly smooth with respect to nuclear motion [8]. Changes in surface elements can cause discontinuities and spurious gradient singularities in the solvation potential, compromising the accuracy of self-consistent field (SCF) energy analytical derivatives [8–10]. Consequently, the previous HSM based on the point-charge PCM scheme, as denoted by PC-HSM, occasionally exhibits unphysical imaginary frequencies due to the discontinuous potential surface and its gradients inherent to the scheme.

To resolve the HSM issues arising from the point-charge scheme, the implementation of an alternative discretization scheme is required. The Gaussian-charge scheme proposed by York and Karplus [8] provides one such approach. In the Gaussian-charge scheme, the surface charge are represented by spherical Gaussian functions centered at the point charge coordinates. Although many Gaussian-charge schemes have been proposed [10–12], they typically combine Gaussian charge smearing with a switching function that damps the charge contribution. This damping prevents discontinuities in the solvation potential when charges effectively appear or disappear during geometry optimization.

In this study, we propose an alternative HSM, referred to as the Gaussian-charge HSM (GC-HSM), which employs the Gaussian charge discretization scheme for PCM. In this Letter, we derive and implement analytical first and second derivatives of the solvation energy within the GC-HSM framework, and report numerical validations of the resulting implementation. In Section 2, the theoretical aspects of the conductor-like PCM (CPCM) equations, the discretization scheme, analytical first and second derivatives, and the HSM scheme are explained. Section 3 describes the computational details. In Section 4, results and discussion are presented, including assessments of the GC-HSM accuracy. Illustrative application to the evaluation of the NaCl dissociation in aqueous solution, numerical

stability of the analytical Hessian matrix, and computational efficiency are also shown. Finally, conclusions are given in Section 5.

2. Theoretical aspects

2.1. *CPCM equations*

For a solvated system within PCM framework, the Hamiltonian of the solute in vacuum, H_0, is perturbed by a solvation term that depends on N_q charges q spread over the PCM cavity surface,

$$H = H_0 + \sum_{i}^{N_q} \frac{q_i}{|\boldsymbol{r}_i - \boldsymbol{r}|}, \tag{1}$$

with $\boldsymbol{r}_i$ denoting the position of the i-th point charge. Within CPCM [13], the charge vector $\boldsymbol{q}$ is given by the following linear equation [14]:

$$\boldsymbol{q} = \boldsymbol{Q}\boldsymbol{v} \quad \text{with} \quad \boldsymbol{Q} = \boldsymbol{K}^{-1}\boldsymbol{R}. \tag{2}$$

Here, $\boldsymbol{v}$ is the electrostatic potential at the position of the charges, and $\boldsymbol{R}$ is the response matrix. The electrostatic potential is given by

$$v_i = \sum_{j=1}^{N_{\text{atoms}}} \frac{Z_j}{|\boldsymbol{r}_i - \boldsymbol{R}_j|} - \sum_{\mu\nu} P_{\mu\nu} \left\langle \chi_\mu \left| \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}|} \right| \chi_\nu \right\rangle. \tag{3}$$

Here, $\boldsymbol{R}_j$ is the j-th atomic nuclear coordinate, Z_j is the nuclear charge, and $P_{\mu\nu}$ is the density matrix element for the basis set $\{\chi_\mu, \chi_\nu\}$. From Eq. (3), the electrostatic potential can be separated into atomic nuclear and electrostatic contributions.

The form of the matrices $\boldsymbol{K}$ and $\boldsymbol{R}$ depend on the type of PCM. In CPCM framework [12], these matrices are given by

$$\boldsymbol{K} = \boldsymbol{A} \tag{4}$$

and

$$\boldsymbol{R} = -\left(\frac{\varepsilon - 1}{\varepsilon + x}\right)\boldsymbol{I} \equiv -f(\varepsilon) \tag{5}$$

Here, ε is the dielectric permittivity of the solvent and $x = 0$ for CPCM [8]. The matrix $\boldsymbol{A}$ is the Coulomb interaction matrix between the solute and the charges on the cavity surface.

The form of the $\boldsymbol{A}$ matrix depends on the discretization scheme and will be discussed in section 2.2. In CPCM, the electrostatic potential due to the point charge on the cavity surface cancels the electrostatic potential of the solute on the cavity surface. Therefore, the surface charge vector $\boldsymbol{q}$ is given by

$$\boldsymbol{A}\boldsymbol{q} = -f(\varepsilon)\boldsymbol{v}\,. \tag{6}$$

Then, the electrostatic solvation energy E_{pol} can be calculated using one of the equivalent forms:

$$E_{\mathrm{pol}} = \frac{1}{2}\boldsymbol{q}^{\mathrm{T}}\boldsymbol{v} = -\frac{1}{2}\boldsymbol{q}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{q}\,. \tag{7}$$

2.2. Discretization scheme

The form of the matrix $\boldsymbol{A}$ is defined by the choice of the discretization scheme, which defines how the surface charge density is represented. There are many choices, however, the Gaussian-charge scheme [8] is widely used because of its robustness and

smoothness of the potential energy surface (PES) compared to the point-charge scheme [2].

In the point-charge scheme, the surface charge vector $\boldsymbol{q}$ given by

$$A_{mn} = \frac{1}{a_m a_n} \int_{a_m} \int_{a_n} \frac{1}{|\boldsymbol{r} - \boldsymbol{r}'|} \mathrm{d}\boldsymbol{r}\mathrm{d}\boldsymbol{r}' . \tag{7}$$

Here, a_m and a_n are the areas of the tesserae m and n. Therefore, each element of the matrix $\boldsymbol{A}$ depends on the tesserae properties such as the position of their centers $\boldsymbol{r}$ and their area, and is given by

$$A_{mn} = 1.07\sqrt{\frac{4\pi}{a_m}} \quad (m = n) \tag{8}$$

and

$$A_{mn} = \frac{1}{r_{mn}} \quad (m \neq n) . \tag{9}$$

The diagonal element in Eq. (8) represents the self-interaction of a point charge, while the off-diagonal element in Eq. (9) represents the inverse distance between r_m and r_n. The off-diagonal element A_{mn} diverges in the limit where two tesserae become infinitely close,

$$\lim_{r_{mn} \to 0} A_{mn} \to \infty . \tag{10}$$

In the Gaussian-charge scheme, the charge density $g_m(\boldsymbol{r})$ is expressed using spherical Gaussian functions centered at the positions $\boldsymbol{r}_m$ on the cavity and is given by

$$g_m(\boldsymbol{r}) = q_m \left(\frac{\zeta_m}{\pi}\right)^{\frac{3}{2}} \exp\left(-\zeta_m^2 (\boldsymbol{r} - \boldsymbol{r}_m)^2\right) \tag{11}$$

Here , ζ_m is the width of the Gaussian function and is given by

$$\zeta_m = \frac{\zeta}{R_I \sqrt{\omega_m}} \tag{12}$$

where ω_m is the Lebedev weight and R_I the radius of the sphere on which the charge is placed. The dimensionless parameter ζ depends on the Lebedev grid and is adjusted to obtain the exact Born solvation energy [8]. Each element of the matrix $\boldsymbol{A}$ is given by

$$A_{mn} = \sqrt{\frac{2}{\pi}} \frac{\zeta_m}{F_m} \quad (m = n) \tag{13}$$

and

$$A_{mn} = \frac{\mathrm{erf}(\zeta_{mn} r_{mn})}{r_{mn}} \quad (m \neq n). \tag{14}$$

with $\zeta_{mn} = \zeta_m \zeta_n / \sqrt{\zeta_m^2 + \zeta_n^2}$. The diagonal element contains the switching function F_m, which is needed to produce a smooth potential surface. F_m is constructed as a product of elementary switching functions $f(\boldsymbol{r}_m, \boldsymbol{R}_I)$ and is given by

$$F_m = \prod_{I, m \notin I}^{N_{\mathrm{atom}}} f(\boldsymbol{r}_m, \boldsymbol{R}_I), \tag{15}$$

where $\boldsymbol{R}_I$ is the position vector of atom I. The off-diagonal element A_{mn} remains finite even when two surface elements become infinitely close,

$$\lim_{r_{mn}\to 0} A_{mn} \to \frac{2\zeta_{mn}}{\sqrt{\pi}}. \tag{16}$$

From the above, the Gaussian-charge scheme avoids discontinuities in the potential energy surface.

2.3. *Analytical first and second derivatives*

The analytical first derivatives of the solvation energy E_{pol} in Eq. (7) with respect to the nuclear coordinates are given by

$$\frac{\partial E_{\mathrm{pol}}}{\partial \boldsymbol{R}_A} = \boldsymbol{q}^{\mathrm{T}} \frac{\partial \boldsymbol{v}}{\partial \boldsymbol{R}_A} + \frac{1}{2f(\varepsilon)} \boldsymbol{q}^{\mathrm{T}} \frac{\partial \boldsymbol{A}}{\partial \boldsymbol{R}_A} \boldsymbol{q}. \tag{17}$$

Here, the derivative of the electrostatic potential $\boldsymbol{v}$ with respect to the nuclear coordinates is given by

$$\frac{\partial \boldsymbol{v}_i}{\partial \boldsymbol{R}_A} = -\sum_{j=1}^{N_{\mathrm{atoms}}} \frac{\partial}{\partial \boldsymbol{R}_A} \frac{Z_j}{|\boldsymbol{r}_i - \boldsymbol{R}_j|} - \sum_{\mu\nu} P_{\mu\nu} \frac{\partial \boldsymbol{v}_i}{\partial \boldsymbol{R}_A} \left\langle \chi_\mu \left| \frac{1}{|\boldsymbol{r}_i - \boldsymbol{r}|} \right| \chi_\nu \right\rangle. \tag{18}$$

Since the derivative of density matrix P with respect to the nuclear coordinates is already included in the self-consistent field (SCF) energy gradient, Eq. (18) does not involve the calculation of this derivative.

The derivative of each element of the matrix $\boldsymbol{A}$ is given by

$$\frac{\partial \boldsymbol{A}_{mm}}{\partial \boldsymbol{R}_A} = \frac{\partial A_{mm}}{\partial \zeta_m} \frac{\partial \zeta_m}{\partial R_A} + \frac{\partial A_{mm}}{\partial F_m} \frac{\partial F_m}{\partial R_A} \tag{19}$$

and

$$\frac{\partial \boldsymbol{A}_{mn}}{\partial \boldsymbol{R}_A} = \frac{\partial A_{mn}}{\partial \zeta_{mn}} \frac{\partial \zeta_{mn}}{\partial R_A} + \frac{\partial A_{mn}}{\partial |\boldsymbol{r}_{mn}|} \frac{\partial |\boldsymbol{r}_{mn}|}{\partial R_A} . \tag{20}$$

When constructing the grid with a fixed cavity radius and Lebedev grid points, the Gaussian width parameter does not depend on the molecular structure, so its derivative including ζ is zero. From Eqs. (13), (14), (19), and (20), each element of the matrix $\boldsymbol{A}$ an be written as

$$\frac{\partial \boldsymbol{A}_{mm}}{\partial \boldsymbol{R}_A} = -\frac{A_{mm}}{F_m} \frac{\partial F_m}{\partial \boldsymbol{R}_A} . \tag{21}$$

and

$$\frac{\partial \boldsymbol{A}_{mn}}{\partial \boldsymbol{R}_A} = \left\{ \frac{2}{\sqrt{\pi}} \frac{\zeta_{mn}}{|\boldsymbol{r}_{mn}|} \exp\left(-\zeta_{mn}^2\right) - \frac{\operatorname{erf}\left(\zeta_{mn} r_{mn}\right)}{|\boldsymbol{r}_{mn}|^2} \right\} \frac{\partial |\boldsymbol{r}_{mn}|}{\partial \boldsymbol{R}_A} . \tag{22}$$

Owing to the smooth representation of the surface charge density in the Gaussian-charge scheme, the solvation potential becomes differentiable with respect to nuclear coordinates. Consequently, analytical second derivatives can be rigorously defined.

The analytical second derivatives of the solvation energy E_{pol} in Eq. (7) with respect to the nuclear coordinates $\boldsymbol{R}_A$ and $\boldsymbol{R}_B$ are given by

$$\begin{aligned} \frac{\partial^2 E_{\mathrm{pol}}}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} = \boldsymbol{q}^{\mathrm{T}} \frac{\partial^2 \boldsymbol{v}}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} + \left[\left(\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{R}_A} \right)^{\mathrm{T}} + \frac{1}{f(\varepsilon)} \boldsymbol{q}^{\mathrm{T}} \frac{\partial \boldsymbol{A}}{\partial \boldsymbol{R}_A} \right] \frac{\partial \boldsymbol{q}}{\partial \boldsymbol{R}_B} \\ + \frac{1}{2 f(\varepsilon)} \boldsymbol{q}^{\mathrm{T}} \frac{\partial \boldsymbol{A}}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} \boldsymbol{q} \end{aligned} . \tag{23}$$

Here, the second derivative of the electrostatic potential $\boldsymbol{v}$ with respect to the nuclear coordinates is given by

$$\frac{\partial \boldsymbol{v}_i}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} = -\sum_{j=1}^{N_{\text{atoms}}} \frac{\partial}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} \frac{Z_j}{\left|\boldsymbol{r}_i - \boldsymbol{R}_j\right|} - \sum_{\mu\nu} P_{\mu\nu} \frac{\partial \boldsymbol{v}_i}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} \left\langle \chi_\mu \left| \frac{1}{\left|\boldsymbol{r}_i - \boldsymbol{r}\right|} \right| \chi_\nu \right\rangle. \tag{24}$$

The second derivative of the surface charge vector $\boldsymbol{q}$ with respect to the nuclear coordinates is given by

$$\frac{\partial \boldsymbol{q}}{\partial \boldsymbol{R}_B} = -\boldsymbol{A}^{-1} \left[\frac{\partial \boldsymbol{A}}{\partial \boldsymbol{R}_B} \boldsymbol{q} + f(\varepsilon) \frac{\partial \boldsymbol{v}}{\partial \boldsymbol{R}_B} \right]. \tag{25}$$

The second derivative of each element of the matrix $\boldsymbol{A}$ is given by

$$\frac{\partial^2 A_{mm}}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} = 2 \frac{A_{mm}}{F_m^2} \frac{\partial F_m}{\partial \boldsymbol{R}_A} \frac{\partial F_m}{\partial \boldsymbol{R}_B} - \frac{A_{mm}}{F_m} \frac{\partial^2 F_m}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} \tag{26}$$

and

$$\begin{aligned} \frac{\partial^2 A_{mn}}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} = & \left[-\frac{2}{\left|r_{mn}\right|} \left\{ \frac{2\zeta_{mn}}{\sqrt{\pi}\left|\boldsymbol{r}_{mn}\right|} \exp\left(-\zeta_{mn}^2 \left|\boldsymbol{r}_{mn}\right|^2\right) \right\} \right] \frac{\partial \left|\boldsymbol{r}_{mn}\right|}{\partial \boldsymbol{R}_A} \frac{\partial \left|\boldsymbol{r}_{mn}\right|}{\partial \boldsymbol{R}_B} \\ & - \frac{\operatorname{erf}\left(\zeta_{mn}\left|\boldsymbol{r}_{mn}\right|\right)}{\left|\boldsymbol{r}_{mn}\right|^2} \frac{\partial \left|\boldsymbol{r}_{mn}\right|}{\partial \boldsymbol{R}_A} \frac{\partial \left|\boldsymbol{r}_{mn}\right|}{\partial \boldsymbol{R}_B} \\ & - \frac{4\zeta_{mn}^3}{\sqrt{\pi}} \exp\left(-\zeta_{mn}^2 \left|\boldsymbol{r}_{mn}\right|^2\right) \frac{\partial \left|\boldsymbol{r}_{mn}\right|}{\partial R_A} \frac{\partial \left|\boldsymbol{r}_{mn}\right|}{\partial R_B} \\ & + \left\{ \frac{2\zeta_{mn}}{\sqrt{\pi}\left|\boldsymbol{r}_{mn}\right|} \exp\left(-\zeta_{mn}^2 \left|\boldsymbol{r}_{mn}\right|^2\right) - \frac{\operatorname{erf}\left(\zeta_{mn}\left|\boldsymbol{r}_{mn}\right|\right)}{\left|\boldsymbol{r}_{mn}\right|^2} \right\} \frac{\partial^2 \left|\boldsymbol{r}_{mn}\right|}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} \end{aligned} \tag{27}$$

Therefore, these derivatives can be calculated by defining the derivatives of the switching function and the distance between grid points with respect to the atomic coordinates.

2.4. *HSM method*

The HSM method explicitly treats the six lowest modes corresponding to the quasi-translational and quasi-rotational motions that are influenced by solute-solvent interactions as harmonic oscillators [1]. Therefore, the expressions for the translational and rotational entropies and enthalpies take the same form as those for vibrational modes, and are given by

$$S_{trans/rot}^{\mathrm{HSM}} = N_A k_{\mathrm{B}} \sum_{i}^{3} \left[\frac{h\nu_i}{k_{\mathrm{B}}T} \frac{1}{\exp\left(\frac{h\nu_i}{k_{\mathrm{B}}T}\right)-1} - \ln\left\{1-\exp\left(-\frac{h\nu_i}{k_{\mathrm{B}}T}\right)\right\} \right] \tag{28}$$

and

$$E_{trans/rot}^{\mathrm{HSM}} = \frac{1}{2} N_A k_{\mathrm{B}} T \sum_{i \in trans/rot}^{3} \frac{h\nu_i}{k_{\mathrm{B}}T} \left\{ \frac{1}{2} + \frac{1}{\exp\left(\frac{h\nu_i}{k_{\mathrm{B}}T}\right)-1} \right\}. \tag{29}$$

Here, h, N_{A}, and k_{B} are the Planck, Avogadro, and Boltzmann constants, respectively. T is the temperature, and $\{\nu_i\}$ are the vibrational frequencies of the normal modes.

In the HSM algorithm, all frequencies are obtained by directly diagonalizing the $3N$-dimensional Hessian matrix because the translational and rotational states are considered on the same footing as the vibrational states.

In the HSM framework, the cavity sphere is treated as a wall with infinite mass. In the differentiation procedure, the position at which the cavity sphere is generated does not follow the change in nuclear coordinates. From eq. (7), the matrix $\boldsymbol{A}$ includes the contribution of the area of tesserae. Since the coordinates of these areas do not move, the first and second derivatives of the matrix $\boldsymbol{A}$ with respect to the nuclear coordinates are zero and given by

$$\frac{\partial E_{\mathrm{pol}}^{\mathrm{HSM}}}{\partial \boldsymbol{R}_A} = \boldsymbol{q}^{\mathrm{T}} \frac{\partial \boldsymbol{v}}{\partial \boldsymbol{R}_A}. \tag{30}$$

and

$$\frac{\partial^2 E_{\mathrm{pol}}}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} = \boldsymbol{q}^{\mathrm{T}} \frac{\partial^2 \boldsymbol{v}}{\partial \boldsymbol{R}_A \partial \boldsymbol{R}_B} - f(\varepsilon) \left(\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{R}_A} \right)^{\mathrm{T}} \boldsymbol{A}^{-1} \frac{\partial \boldsymbol{v}}{\partial \boldsymbol{R}_B}. \tag{31}$$

In this study, we implemented an alternative HSM referred to as the Gaussian-charge HSM (GC-HSM). GC-HSM is based on the PCM of Gaussian-charge scheme, so, it is expected that continuous PES can be obtained.

3. Computational details

In this section, we present illustrative applications of PC-HSM and GC-HSM based on the point-charge and Gaussian-charge schemes, respectively. We have

implemented the GC-HSM in PySCF (ver. 2.11.0) [15]. The PC-HSM calculations were performed using the GAMESS 2022R1 [16] program .

Geometry optimizations were carried out at the Hartree-Fock [17,18] level with a cc-pVDZ basis set [19]. Solvent effects were treated by CPCM [13] with water (ε = 80.1510). The solute cavity was constructed from Bondi's van der Waals radii [20] scaled by 1.2 for all atoms, except for hydrogen atoms for which a scaling factor of 1.1 [21] was employed.

The cavity surface discretization used 60 tesserae per atom for PC-HSM and a Lebedev grid of 110 points per atom for GC-HSM. The numerical Hessian matrix for the HSM was evaluated by a second-order accurate central-difference scheme of analytical gradients, requiring $6N + 1$ gradient evaluations. For timing benchmarks in Section 4.3, the displacement was 0.01 Bohr. All calculations were performed using default integral thresholds unless otherwise specified. Timing tests were performed on Research Center for Computational Science using a single node of AMD EPYC 7763 (8 cores).

4. Results and discussion

4.1. Dissociation of NaCl in aqueous solution

The smoothness of the PES and its gradient along the dissociation of NaCl in aqueous solution was examined. Figures 2 and 3 show PESs of the solvation energy E_{pol} defined in Eq. (7), and its analytical gradient with respect to the displacement of the Cl atom, using PC-HSM and GC-HSM, respectively. In both figures, panel (a) displays the PES for Na-Cl distances in the range of 2.0 to 5.0 Å, and panel (b) displays the PES in the range of 2.993 to 2.994 Å. As shown in Fig. 2(a), the energy curve (red line) and its gradient (blue line) using PC-HSM appeared smooth. In contrast, as shown in Fig. 2(b), the energy curve exhibited step-like discontinuities when examined at 0.001 Å intervals, whereas a continuously varying analytical gradient was obtained.

The inconsistency between the energy and its first derivative originates from the formulation of the gradient defined in Eq. (30). In the point-charge scheme, the definition of matrix $\boldsymbol{A}$ leads to discontinuities in the PES [8]. This behavior indicates that the PES obtained with the point-charge discretization is not strictly differentiable with respect to nuclear coordinates. However, in the HSM framework, the form of the analytical first derivative of solvation energy ignores the contribution of the matrix derivative and therefore fails to reflect the discontinuous jumps in the PES.

For the GC-HSM shown in Fig. 3, the energy curve remained continuous even in the narrow range as shown in Fig. 3(b). This result indicates that the Gaussian

distribution of the cavity surface charges eliminated the potential surface discontinuity associated with the point-charge scheme.

Although minor oscillations in the gradient were observed at long distances above 4.0 Å in Fig. 3(a), these are fundamentally different from the discontinuities of the PC-HSM. As suggested in ref. [8], these artifacts originate from numerical integration errors on the finite grid and can be reduced by increasing the grid density.

4.2 Numerical stability of Hessian matrix

Next, the accuracy of the analytical Hessian matrix defined in the HSM framework is examined. As shown in section 4.1, the PC-HSM exhibits an inconsistency where a continuous analytical gradient is obtained despite the discontinuous PES. The inconsistency implies that a rigorous analytical Hessian matrix cannot be defined in the PC-HSM. Therefore, this section focuses on the validation of the analytical Hessian matrix for the GC-HSM, which enables the analytical second derivative to be rigorously defined due to the smoothness of the PES. The numerical stability of the analytical Hessian matrix was examined using a water molecule. To confirm the correctness of the analytical second derivatives, comparisons with numerical finite-difference Hessians were performed.

Figure 4 shows the absolute errors in the difference between the Frobenius norms of the analytical ($\boldsymbol{H}_{\text{ana}}$) and numerical ($\boldsymbol{H}_{\text{num}}$) Hessian matrices, denoted as $\left| \|\boldsymbol{H}_{\text{ana}}\|_F - \|\boldsymbol{H}_{\text{ana}}\|_F \right|$, plotted against the finite difference step size. The inset the absolute errors of the individual upper triangular elements of the Hessian submatrix $\boldsymbol{H}$ corresponding to the oxygen atomic coordinates. The color of each plot indicates the corresponding element of the Hessian matrix.

Large deviations were observed in the region of small step sizes smaller than 0.003 Bohr. These errors were attributable to the loss of significant digits inherent to the finite difference method. In contrast, for step size larger than 0.005 Bohr, the difference in the Frobenius norm decreases to less than 0.001 kcal mol^{-1} Bohr^{-2}. This agreement verifies the correctness of the analytical second-derivative form defined in the HSM framework and its implementation. Therefore, the GC-HSM yields a numerically stable Hessian matrix consistent with the continuous PES.

4.3. Computational efficiency

Finally, the computational efficiency of the analytical Hessian matrix was assessed using linear alkane molecules, C_nH_{2n+2} (n = 1, 2, …, 10). The computational advantage of analytical second derivatives over numerical differentiation was also

investigated. Figure 5 compares the wall-clock times required for the Hessian calculations using numerical differentiation and analytical differentiation implemented in the GC-HSM. The computational time for numerical differentiation (blue circles) increased drastically with the number of carbon atoms. For example, the Hessian calculation for the decane (n = 10) took approximately 68000 seconds, while the analytical Hessian (red diamond) took approximately 6900 seconds. As a result, the analytical differentiation achieved approximately a tenfold speedup over numerical differentiation for a decane molecule. These results demonstrate that the implementation of analytical second derivatives in the GC-HSM significantly reduces computational cost, thereby extending the applicability of the HSM to large-scale systems.

5. Conclusions

In this study, the analytical first and second derivatives of the solvation energy was formulated and implemented for GC-HSM within the HSM framework. The performance of the GC-HSM was assessed by comparing it with the previous HSM (PC-HSM) and with results obtained by numerical differentiation. The investigation of the PES for Na⁻Cl dissociation revealed that the PC-HSM exhibited step-like discontinuities and an inconsistency between the energy and its analytical gradient. In contrast, the GC-

HSM yielded a smooth PES and a consistent analytical gradient, thereby allowing analytical second derivative to be rigorously defined. Numerical validations for a water molecule confirmed that the analytical Hessian matrix of the GC-HSM is numerically stable and accurate, and showed excellent agreement with the results of finite-difference calculations. The computational efficiency tests using linear alkane chains demonstrated that the analytical Hessian matrix significantly reduces the computational cost compared to numerical differentiation. The present formulation enables fully analytical Hessian calculations within the HSM framework for solvated systems. These results suggest that GC-HSM overcomes the limitations of PC-HSM and provides efficient and reliable results for large systems in the condensed phase.

CRediT authorship contribution statement

Takuya Hashimoto: Writing – original draft, Visualization, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hiromi Nakai:** Writing – review & editing, Validation, Supervision, Software, Resources, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Figure Captions

Figure 1. Definition of cavity spheres generated by (a) the point-charge scheme and (b) the Gaussian-charge scheme.

Figure 2. Potential energy curves and analytical gradients for the dissociation of NaCl in aqueous solution calculated using the PC-HSM. Profile over the Na-Cl distance range of (a) 2.0 to 5.0 Å and (b) 2.993 to 2.994 Å. The red and blue lines correspond to the solvation energy and its analytical gradient with respect to the displacement of the Cl atom, respectively.

Figure 3. Potential energy curves and analytical gradients for the dissociation of NaCl in aqueous solution calculated using the GC-HSM. Profile over the Na-Cl distance range of (a) 2.0 to 5.0 Å and (b) 2.993 to 2.994 Å. The red and blue lines correspond to the solvation energy and its analytical gradient with respect to the displacement of the Cl atom,

Figure 4. Numerical stability of the analytical Hessian for a water molecule calculated using the GC-HSM. The main panel shows the Frobenius norm of the difference matrix

between the analytical and numerical Hessian matrices as a function of the finite-difference step size. The inset panel displays the absolute errors of the individual upper triangular elements of the subhessian block corresponding to the oxygen atom coordinates.

Figure 5. Computational efficiency of Hessian calculations for linear alkane chains, C_nH_{2n+2}. The wall-clock times required for analytical differentiation (red diamonds) and numerical differentiation (blue circles) are plotted against the number of carbon atoms.

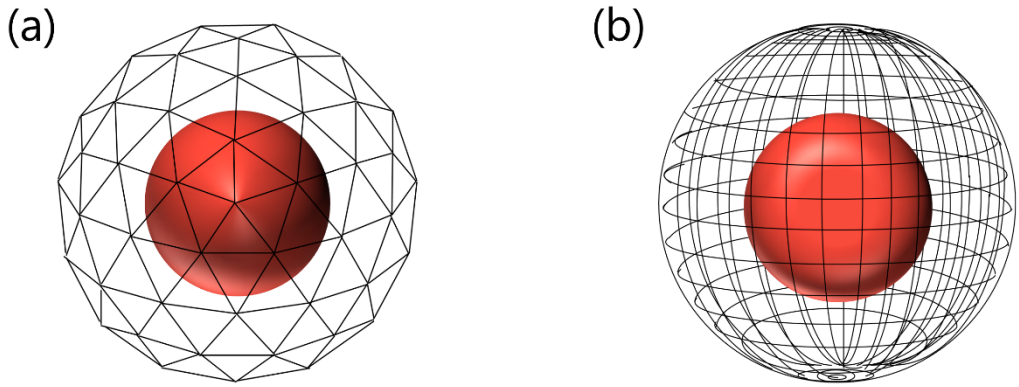
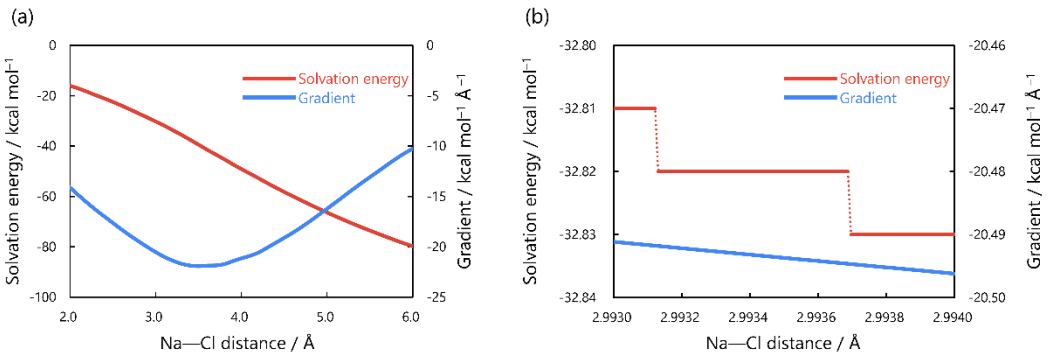


Figure 1 of Hashimoto and Nakai



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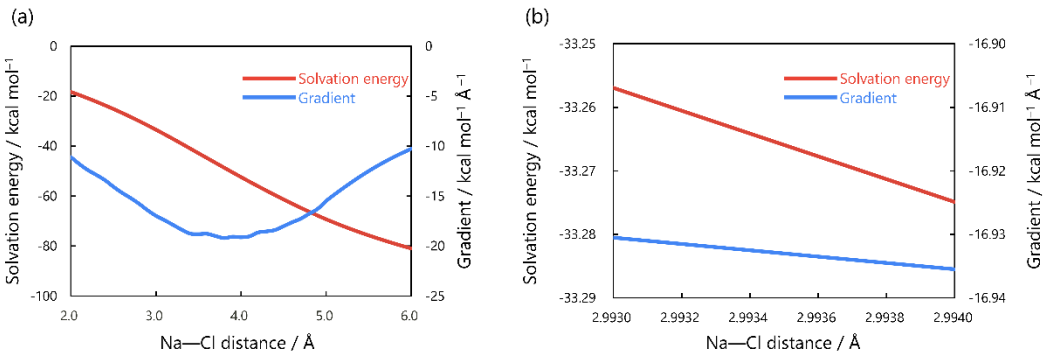

Figure 3 of Hashimoto and Nakai

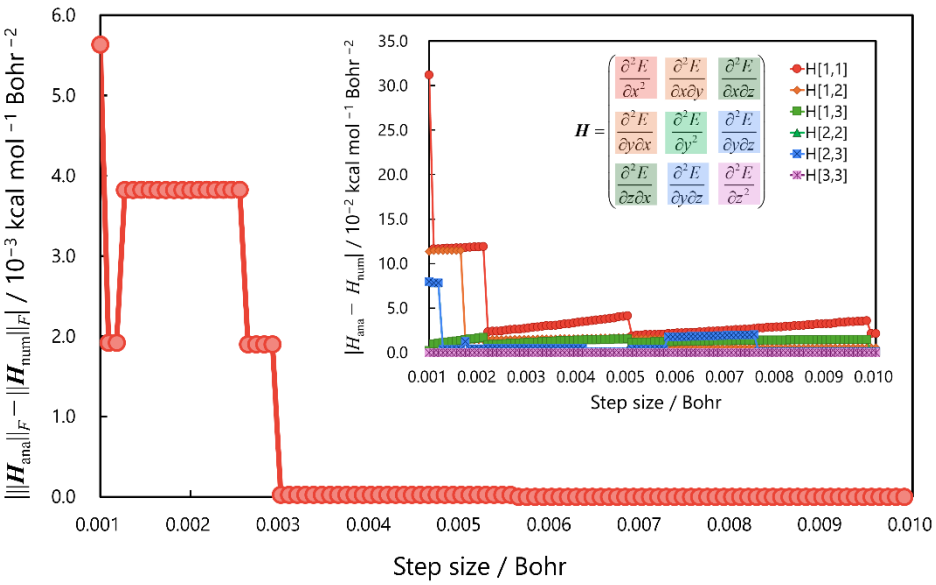

Figure 4 of Hashimoto and Nakai

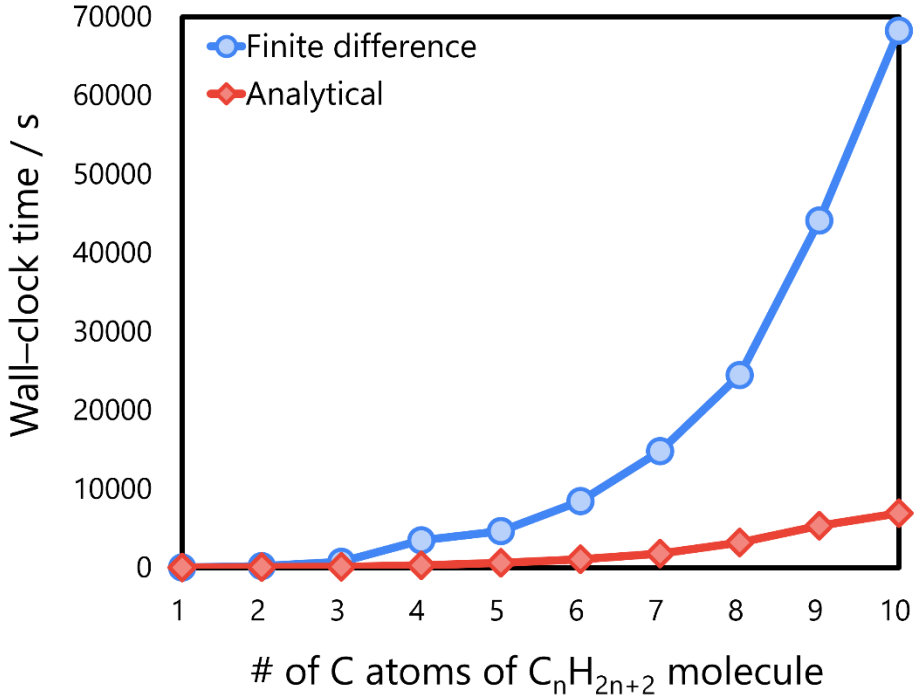

Figure 5 of Hashimoto and Nakai